

Principles of Communication Systems

Mid-Term Examination

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Duration: 90 minutes Total: 20 marks

Please provide your answers clearly, with all the steps outlined. Some of the trigonometric identities required are as follows:

- $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$
- $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$
- $\cos(\theta_1 - \theta_2) = \cos(\theta_1)\cos(\theta_2) + \sin(\theta_1)\sin(\theta_2)$
- $\cos(\theta_1 + \theta_2) = \cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2)$

1. Part A

Problem 1 Consider a modulating signal $m(t)$ has zero mean and maximum value of $A_m = \max[m(t)]$. The carrier waveform is given by $c(t) = \cos 2\pi f_c t$. The non-linear device has input output relationship given by $y(t) = ax(t) + bx^2(t)$ where $x(t) = m(t) + c(t)$. Can the system specified above generate a conventional amplitude modulated signal? If yes, what is the modulation index? Specify the nature of the filter that needs to be used after the square law device to obtain the conventional AM signal at the output. (4 marks)

Problem 2 An analog baseband message signal $m(t)$ that entirely lies within $[-W, W]$ Hz is up-converted to a passband signal, $v_p(t)$ where

$$v_p(t) = m(t)\sin(2\pi f_c t + \Phi_c),$$

with $f_c > 10W$. The channel delays the signal such that the received signal is given by

$$r(t) = v_p(t - d).$$

At the receiver, the mixer output is given by

$$y(t) = r(t)\sin(2\pi f_c t + \Phi_r).$$

1. Rewrite $y(t)$ in terms of f_c , $\Phi_c - \Phi_r$ and d ... (2)
 2. Let $v(t)$ be obtained by low-pass filtering $y(t)$. Write the expression for $v(t)$ and mention the cut-off frequency for such a low pass filter if $v(t)$ needs to be in baseband... (0.5)
 3. Let $d = nT_c + T_c/4$, where n is an integer. What is the minimum and maximum value of $v(t)$ if $\Phi_c = \Phi_r$ and $T_c = 1/f_c$... (1.5)
- .. (4)

Problem 3 Let the carrier waveform be given by $c(t) = 10\cos(2\pi f_c t)$ and message signal by given by $m(t) = \cos(20\pi t)$. The message signal is used to frequency modulate the carrier with $k_f = 50$. Answer the following: (5).

- (a) Find the expression for the modulated wave ($u(t)$).
- (b) What is the power and modulation ratio of $u(t)$?

- (c) Use Carson's rule to provide an estimate of the bandwidth of $u(t)$.

Problem 4 A superheterodyne receiver is designed to receive signals with carrier frequencies between 4 and 6 MHz, with transmitted bandwidths of 100KHz each. The intermediate frequency is 850KHz .

- (a) What is the 'range' of local oscillator frequency if $f_{LO} > f_c$ is used? .. (1)
 (b) What is the range of image frequencies? .. (1)

2. Part B

Problem 1 Consider a signal $x(t) \leftrightarrow X(f)$ whose sampled form is given by $\sum_{-\infty}^{\infty} x(nT)\delta(t-nT)$, and the Nyquist sampling rate is $f_s = 2B$ where B is the bandwidth $x(t)$. Find Nyquist sampling frequency for the signal $y(t) \leftrightarrow Y(f)$, where:

- (a) If $\text{Re}(X'(f)) = \text{Re}(X(f)) \dots (1)$
 (b) $y(t) = x(t)\cos(2\pi f_s t) \dots (1)$
 (c) $y(t) = x(t/2) \dots (1)$
 (d) $y(t) = x(t-d) \dots (1)$
 (e) $y(t) = x(t-\theta(t))$, where $\theta(t)$ is a time-varying phase. ... (1)